# Differentiate exponential functions with base e

**1**

a $y' = 2xe^{x^2+5}$

b $y' = (2x+3)\,e^{x^2+3x+5}$

c $y' = (8x-3)\,e^{4x^2-3x}$

d $y' = 15e^{5x} + 4e^{-4x} + 2x$

e $y' = 3e^{3x} + 10x$

f $y' = x^5 e^{\frac{1}{6}x^6}$

g $y' = 18xe^{3x^2}$

h $y' = 3\,(2x-3)\,e^{x^2-3x+4}$

i $y' = 2e^{\sqrt{x}}x^{-\frac{1}{2}}$

**2**

a $y' = \dfrac{1-x}{e^x}$

b $y' = \dfrac{e^{3x}\,(3x-1)}{5x^2}$

c $y' = \dfrac{1-6x}{2e^{3x}\sqrt{x}}$

d $y' = -2e^{2x} - e^{-x}$

e $y' = \dfrac{e^{6x}\,(6+5e^x)}{(1+e^x)^2}$

f $y' = \dfrac{e^x\left(7-5e^{6x}\right)}{\left(e^{6x}+7\right)^2}$

g $y' = \dfrac{14e^x}{(e^x+7)^2}$

h $y' = \dfrac{-30e^{3x}}{\left(e^{3x}-5\right)^2}$

i $y' = e^{3x-4}\,(x+2)\,(3x+8)$

j $y' = e^{-4x}\left(\dfrac{1}{2\sqrt{x+2}} - 4\sqrt{x+2}\right)$

k $y' = \dfrac{e^{2x}\,(2x-1)}{x^2}$

l $y' = x^4e^{4x}\,(4x+5)$

m $y' = x^4e^{-6x}\,(5-6x)$

n $y' = e^x\left(x^4+4x^3+1\right)$

o $y' = e^{2x}\left(2x^3+3x^2+8x+14\right)$

p $y' = -e^{-x}\left(x^2+6x-6\right)$

q $y' = x\left(3x+3xe^{3x}+2e^{3x}\right)$

**3**

a $y' = 4e^x\,(e^x-4)^3$

b $y' = 7\left(2x-e^{3x}\right)^6\left(2-3e^{3x}\right)$

c $y' = \dfrac{3}{4}\left(e^{2x}+5x\right)^{-\frac{1}{4}}\left(2e^{2x}+5\right)$

d $f'(x) = 5e^x\,(e^x+4)^4$

e $f'(x) = 6\left(2e^{2x}+3\right)\left(e^{2x}+3x\right)^5$

f $f'(x) = 4\left(4e^{4x}+2x\right)\left(e^{4x}+x^2\right)^3$

g $f'(x) = 210e^{6x}\left(5e^{6x}-9\right)^6$

h $f'(x) = -6e^{2x}\left(16-e^{2x}\right)^3$

i $f'(x) = 7\left(5+6e^{6x}\right)\left(5x+e^{6x}\right)^6$

j $f'(x) = 8xe^{x^2}\left(e^{x^2}-8\right)^3$

k $f'(x) = \dfrac{-6e^x}{(e^x+1)^7}$

l $f'(x) = \dfrac{10e^{2x}-2e^{3x}}{(e^x+5)^5}$

**4**

a $f'(x) = 4e^{4x} + 2e^{2x}$

b $f'(0) = 6$

**5** $f'(1) = -16e^{-3}$

## Differentiate exponential functions with other bases

**6**

a $y = e^{(\ln 7)x}$

b $y' = (\ln 7)7^x$

c $7 \ln 7$

**7**

a $f'(x) = (\ln 5)5^x$

b $f'(x) = 2(\ln 7)7^x$

c $f'(x) = -(\ln 4)4^x$

d $f'(x) = \dfrac{3x^2 - (\ln 2)\left(x^3 - 1\right)}{2^x}$

e $f'(x) = 5^x x\,(x \ln 5 + 2)$

f $f'(x) = -(\ln 2)2^{-t}$

g $P'(t) = 3(\ln 2)2^{3t}$

h $f'(x) = -4(\ln 5)5^{2-4x}$

## Applications

**8**

a $8$

b $y = 8x + 7$

**9**

a $-\dfrac{1}{6}$

b $x + 6y - 6 = 0$

**10** $y = 2x - 8$

**11** $x - 4y - 8 = 0$